GATE ME 90 Days Challenge

Day 25/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Machine Design	Joints	82 days to GATE

How to use this set: 15 questions (5 MCQ, 10 NAT) on riveted joints (modes of failure, strength and efficiency, boiler joints), welded joints (butt, transverse and parallel fillet welds, circular welds, eccentrically loaded welds) and bolted joints (preload, joint stiffness, separation, eccentric loading in shear and in tension). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

Caulking and fullering are done on riveted joints of boilers and pressure vessels to

(A) reduce the stress concentration at the rivet holes

(B) allow for corrosion of the plates

(C) increase the shearing strength of the rivets

(D) make the joint leak-proof

Q2 NAT

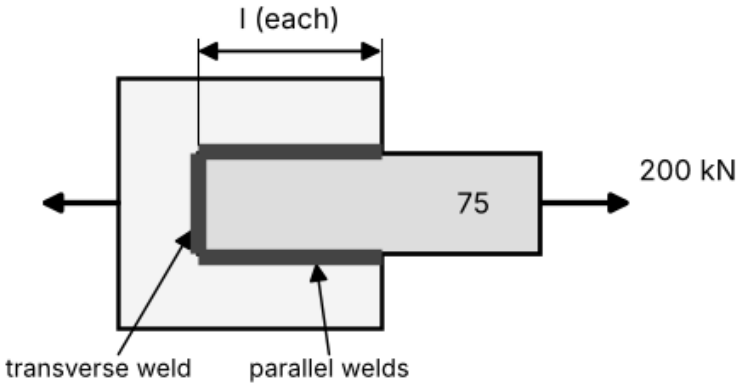
A double-riveted lap joint (two rivets per pitch length, single shear) joins 12 mm plates. Hole diameter = 22 mm, pitch = 75 mm. Permissible stresses: σ_t = 90 MPa, τ = 70 MPa, σ_c = 150 MPa. The efficiency of the joint (in %, round off to one decimal place) is _____

Q3 NAT

A boiler shell of 1200 mm internal diameter is made of 14 mm plates. The permissible tensile stress is 90 MPa and the efficiency of the longitudinal riveted joint is 80 %. The maximum allowable steam pressure (in MPa, round off to two decimal places) is _____

Q4 NAT

A 75 mm wide plate is lap-welded to another plate by a transverse fillet weld across its full width plus two parallel fillet welds of equal length, all of 10 mm leg size, as shown. The joint carries 200 kN. The permissible stresses in the weld are 90 MPa in tension (transverse weld) and 70 MPa in shear (parallel welds). The length of each parallel weld (in mm, round off to one decimal place) is _____



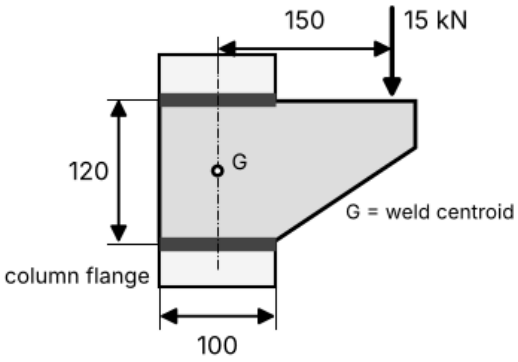
Q5 MCQ

The polar moment of inertia of a straight weld line of length l about its own centre, per unit throat thickness (the 'unit' polar moment), is

(A) $l^3/12$
(B) $l^3/6$
(C) $l^3/3$
(D) $l^2/12$

Q6 NAT

A bracket is welded to a column by two horizontal fillet welds, each 100 mm long and 120 mm apart, as shown. A vertical load of 15 kN acts in the plane of the welds at 150 mm from the centroid of the weld group. For a permissible shear stress of 80 MPa, the leg size of the welds (in mm, round off to two decimal places) is _____



Q7 NAT

An M16 bolt is to be given an initial tension of 30 kN. Taking the torque coefficient as 0.2, the wrench torque required (in N·m) is _____

Q8 NAT

A bolted joint has an initial preload of 30 kN and a joint constant $C = k_b/(k_b + k_m) = 0.25$. The external tensile load at which the clamped members just separate (in kN) is _____

Q9 MCQ

A 'bolt of uniform strength' has its shank turned down to the core diameter of the threads (or a hole drilled along its axis). The main purpose is to

(A) make it easier to tighten

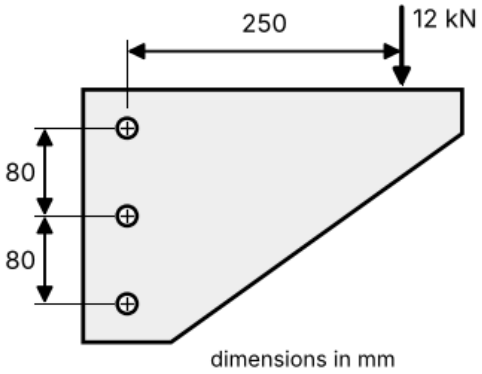
(B) increase its capacity to absorb shock (impact) energy

(C) increase its stiffness

(D) increase its static tensile strength

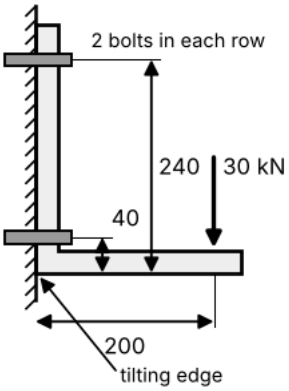
Q10 NAT

A plate is bolted to a column by three bolts in a vertical line at a pitch of 80 mm, as shown. A vertical load of 12 kN acts at a horizontal distance of 250 mm from the middle bolt. The resultant shear force on the most heavily loaded bolt (in kN, round off to two decimal places) is _____



Q11 NAT

A bracket is fixed to a wall by four bolts, two at 40 mm and two at 240 mm from the lower edge about which it tends to tilt, as shown. A load of 30 kN acts at 200 mm from the wall. The maximum tensile load on a bolt due to the overturning moment (in kN, round off to two decimal places) is _____



Q12 MCQ

In designing a single-V butt weld loaded in tension, the strength is calculated as

(A) 1.414 t l σ_t

(B) (t + reinforcement) l σ_t

(C) 0.707 t l σ_t, where t is the plate thickness

(D) t l σ_t, where t is the plate thickness (reinforcement neglected)

Q13 NAT

A 50 mm diameter rod is welded to a plate by a circumferential fillet weld of 6 mm leg size. The rod is subjected to a bending moment of 500 N·m. The maximum bending stress in the weld (in MPa, round off to two decimal places) is _____

Q14 MCQ

In a riveted joint, the margin (distance from the centre of a rivet hole to the edge of the plate) is usually taken as about

(A) 0.5 d
(B) 3 d
(C) 1.5 d
(D) 5 d

Q15 NAT

The cover of a cylinder of 300 mm inside diameter is held by 12 bolts. The gas pressure is 2 MPa and the permissible tensile stress in the bolts is 80 MPa. Neglecting the preload, the minimum core diameter of the bolts (in mm, round off to two decimal places) is _____

Answer Key

Q	Type	Answer
Q1	MCQ	(D) make the joint leak-proof
Q2	NAT	65.7 (accept 65.3 to 66.1)
Q3	NAT	1.68 (accept 1.67 to 1.69)
Q4	NAT	153.8 (accept 152.9 to 154.8)
Q5	MCQ	(A) $l^3/12$
Q6	NAT	4.47 (accept 4.43 to 4.51)
Q7	NAT	96
Q8	NAT	40
Q9	MCQ	(B) increase its capacity to absorb shock (impact) energy
Q10	NAT	19.17 (accept 19.06 to 19.29)
Q11	NAT	12.16 (accept 12.09 to 12.24)
Q12	MCQ	(D) t l σ_t, where t is the plate thickness (reinforcement neglected)
Q13	NAT	60.03 (accept 59.67 to 60.39)
Q14	MCQ	(C) 1.5 d
Q15	NAT	13.69 (accept 13.61 to 13.78)

Detailed Solutions

Q1 **Answer: (D) make the joint leak-proof**

Concept: In caulking, the bevelled plate edge is hammered with a blunt tool so that it presses tightly on the other plate; fullering uses a tool as wide as the plate edge.

Both close the gap between the plates so that steam or fluid cannot leak: the joint becomes **leak-proof (fluid-tight)**. They add no strength.

Exam tip: For caulking, the plate edge is planed at about 75–80° before hammering.

Q2 **Answer: 65.7 (accept 65.3 to 66.1)**

Concept: Work out all three strengths over one pitch length and divide the least by the strength of the solid plate.

Tearing: $P_t = (p - d)t\sigma_t = (75 - 22) \times 12 \times 90 =$ **57240 N**

Shearing: $P_s = 2 \times (\pi/4) \times 22^2 \times 70 =$ **53219 N**

Crushing: $P_c = 2 \times 22 \times 12 \times 150 =$ **79200 N**

Solid plate: p t $\sigma_t = 75 \times 12 \times 90 = 81000$ N → $\eta = 53219/81000 =$ **65.7 %**

Exam tip: Shearing governs here, so increasing the rivet diameter (not the pitch) is the way to raise this joint's efficiency.

Q3 **Answer: 1.68 (accept 1.67 to 1.69)**

Concept: The longitudinal joint resists the hoop stress: $\sigma = pD/(2t\eta)$.

$p = 2t\sigma\eta/D = 2 \times 14 \times 90 \times 0.8/1200 =$ **1.68 MPa**

Exam tip: The circumferential joint carries only the longitudinal stress pD/4t, so it may have about half the efficiency of the longitudinal joint.

Q4 **Answer: 153.8 (accept 152.9 to 154.8)**

Concept: The welds share the load: transverse weld $0.707\ s\ l\ \sigma_t$ + parallel welds $2 \times 0.707\ s\ l\ \tau$.

Transverse weld: $0.707 \times 10 \times 75 \times 90 =$ **47722 N**

Remaining load = $200000 - 47722 = 152278$ N $= 2 \times 0.707 \times 10 \times l \times 70 = 989.8\ l$

$l =$ **153.8 mm**

Exam tip: A single transverse weld across the end of a lap joint is used here; a double transverse lap joint has welds at both plate ends.

Q5 **Answer: (A) $l^3/12$**

Concept: Treat the weld as a line. About its centre, the unit moment of inertia about the perpendicular axis is $\int x^2\,dx$ from $-l/2$ to $l/2$.

$J_u =$ **$l^3/12$**. The actual polar moment is $J = 0.707\ s \times J_u$.

Exam tip: For a weld group, shift each line to the group centroid with $J_u = l^3/12 + l \times (\text{distance})^2$.

Q6 **Answer: 4.47 (accept 4.43 to 4.51)**

Concept: Eccentric load in the plane of the welds = direct shear P/A + torsional shear Pe·r/J. Work per unit throat t, then find t.

Unit polar moment of the two lines about the centroid: $J_u = b(3d^2 + b^2)/6 = 100(3 \times 120^2 + 100^2)/6 =$ **886667 mm³**

Critical point: weld end nearest the load, at x = 50, y = 60 mm (r = 78.1 mm)

Vertical: $P/(2b) + Pe{\cdot}x/J_u = 75 + 126.88 =$ **201.88/t**; horizontal: $Pe{\cdot}y/J_u =$ **152.26/t**

Resultant $= \sqrt{(201.88^2 + 152.26^2)}/t = 252.86/t = 80 \rightarrow t = 3.161$ mm, $s = t/0.707 =$ **4.47 mm**

Exam tip: Add the components vectorially; adding the magnitudes of the direct and torsional shear overestimates the stress.

Q7 **Answer: 96**

Concept: $T = K F_i d$.

$T = 0.2 \times 30000$ N $\times 0.016$ m = **96 N·m**

Exam tip: K ≈ 0.2 for dry, unlubricated threads; lubrication lowers it to about 0.15, so the same torque gives more preload.

Q8 **Answer: 40**

Concept: Member compression $= F_i - (1 - C)P$. Separation when this becomes zero.

$P_{sep} = F_i/(1 - C) = 30/(1 - 0.25) =$ **40 kN**

At that load the bolt carries the whole external load: $F_b = 30 + 0.25 \times 40 = 40$ kN = P.

Exam tip: Beyond separation every extra newton of external load goes into the bolt; a joint must never be designed to separate.

Q9 **Answer: (B) increase its capacity to absorb shock (impact) energy**

Concept: Under impact, energy absorbed = $\sigma^2 V/(2E)$. In a normal bolt the threaded part (smaller area) reaches the limiting stress while the thicker shank is lightly stressed.

Reducing the shank to the core area makes the stress uniform along the length, so the whole volume works at the limiting stress and the bolt **absorbs more shock energy**.

Exam tip: It also lowers the bolt's stiffness, which helps under fluctuating loads, but its static strength (set by the core area) does not change.

Q10 **Answer: 19.17 (accept 19.06 to 19.29)**

Concept: Direct shear P/n on every bolt plus secondary shear $F_i'' = P\, e\, r_i/\Sigma r^2$ perpendicular to each radius.

Direct shear = 12/3 = **4 kN** (vertical)

Secondary on top and bottom bolts (r = 80 mm; middle bolt r = 0): $12 \times 250 \times 80/(2 \times 80^2)$ = **18.75 kN**, horizontal

Resultant = $\sqrt{(4^2 + 18.75^2)}$ = **19.17 kN** (top and bottom bolts)

Exam tip: Here the two components are perpendicular; for bolts arranged in a square they meet at 45°, so use the cosine rule.

Q11 **Answer: 12.16 (accept 12.09 to 12.24)**

Concept: The bracket tends to rotate about the tilting edge; bolt elongation, and hence bolt load, is proportional to the distance from that edge: F = w l.

$P\, L = 2w(l_1^2 + l_2^2)$: $w = 30 \times 200/(2 \times (40^2 + 240^2)) = 0.05068$ kN/mm

$F_{max} = w \times 240$ = **12.16 kN** (the two upper bolts)

Exam tip: Each bolt also carries direct shear 30/4 = 7.5 kN; combine both for the bolt size with a theory of failure.

Q12 **Answer: (D) $t\, l\, \sigma_t$, where t is the plate thickness (reinforcement neglected)**

Concept: In a butt weld the throat equals the thickness of the thinner plate.

$P = \mathbf{t\, l\, \sigma_t}$. The reinforcement (about 10 % of t) is ignored as a margin of safety.

Exam tip: The factor 0.707 belongs to fillet welds, whose throat is 0.707 × leg size.

Q13 **Answer: 60.03 (accept 59.67 to 60.39)**

Concept: Treat the weld as a thin ring of diameter d and thickness t (throat): $Z = \pi\, t\, d^2/4$.

t = 0.707 × 6 = **4.242 mm**

$\sigma_b = M/Z = 4M/(\pi\, t\, d^2) = 4 \times 500000/(\pi \times 4.242 \times 50^2)$ = **60.03 MPa**

Exam tip: Under torsion the same weld gives $\tau = 2T/(\pi\, t\, d^2)$, exactly half of $4T/(\pi\, t\, d^2)$.

Q14 **Answer: (C) 1.5 d**

Concept: The margin must be large enough that the plate does not tear or shear out between the hole and the edge.

Standard practice: m = **1.5 d** (d = rivet hole diameter). The minimum pitch is about 2 d to leave room for the rivet head and tools.

Exam tip: Do not confuse margin (hole centre to plate edge) with pitch (hole centre to hole centre along a row).

Q15 **Answer: 13.69 (accept 13.61 to 13.78)**

Concept: The total gas force on the cover is shared equally by the bolts.

$F = (\pi/4)D^2p = (\pi/4) \times 300^2 \times 2 =$ **141372 N**

Per bolt = 141372/12 = **11781 N**

$d_c = \sqrt{(4 \times 11781/(\pi \times 80))} =$ **13.69 mm**; choose the next standard size (M16, $d_c \approx 13.8$ mm).

Exam tip: Bolts must be sized on the core (root) diameter, which is about 0.84 times the nominal diameter.

What next?

Day 25 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 26/90: Machine Design, Machine Elements. Comment **GATE90** on our reel to get all links. Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.